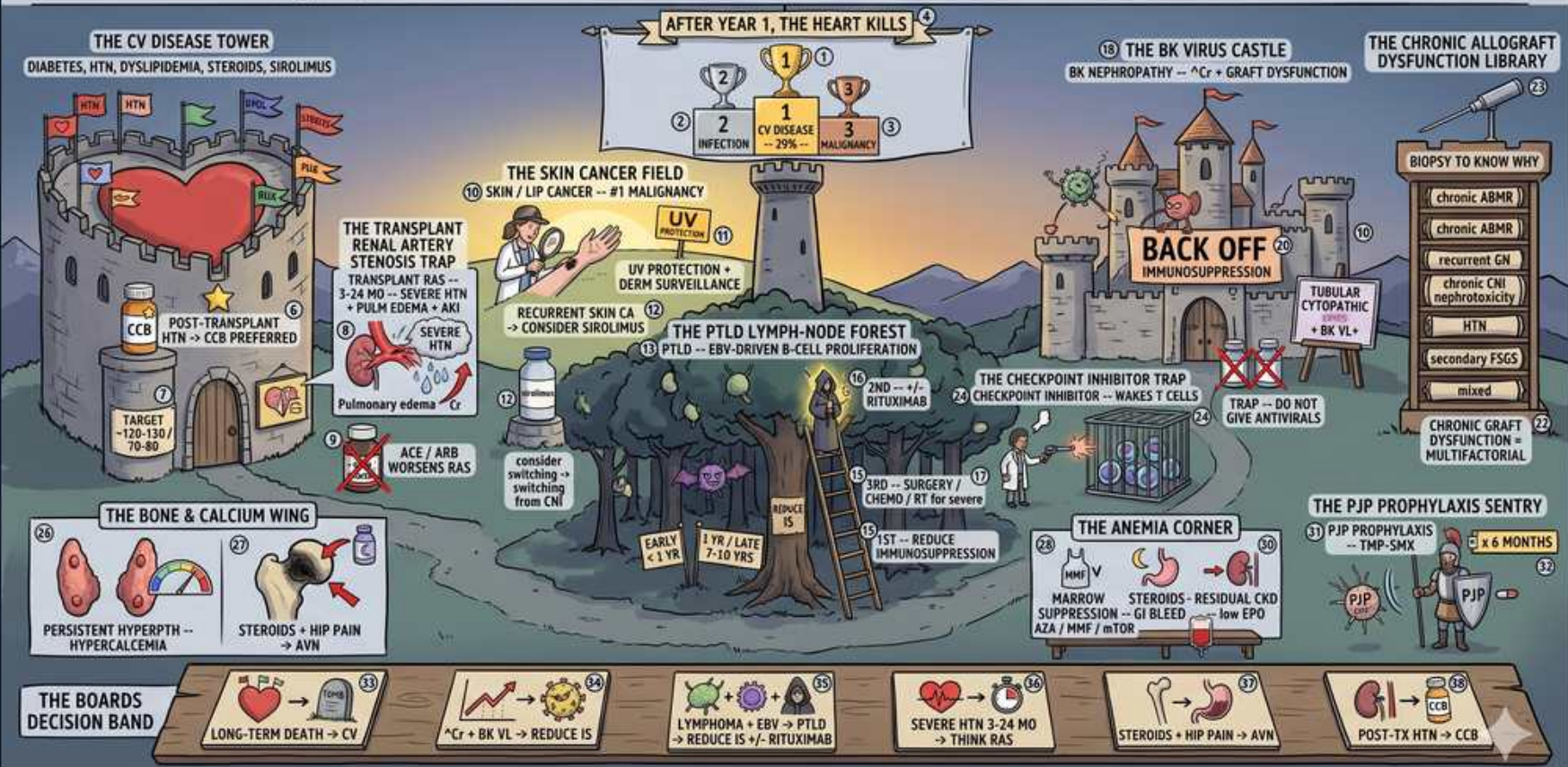


Chronic Complications After Kidney Transplant — The Long Watch

THE LONG WATCH — CHRONIC COMPLICATIONS AFTER KIDNEY TRANSPLANT

After year 1, the heart kills. Skin cancer #1. PTLD = EBV. BK = Back off IS. Severe HTN 3-24mo = think RAS.



After year 1, CV disease kills most. Skin cancer = #1 malignancy. PTLD = EBV-driven, reduce IS first. BK = back off IS, never antivirals. Severe HTN 3-24mo = transplant RAS. PJP prophylaxis = TMP-SMX x 6 months.

1. After year 1, CV disease kills most

WHAT YOU SEE

CV disease tower, post-year-1 banner

WHAT IT ENCODES

Beyond the first year, cardiovascular disease is the LEADING cause of death with a functioning graft. Aggressive management of HTN, dyslipidemia, and diabetes is central to long-term survival.

BOARD TRAP

Late post-transplant death is cardiovascular, not infection — infection dominates the FIRST month-to-year window, not the chronic phase.

2. Post-transplant HTN — CCB preferred

WHAT YOU SEE

CV tower note: HTN target, CCB preferred

WHAT IT ENCODES

Hypertension is common (CNI-driven vasoconstriction). Dihydropyridine calcium channel blockers are often preferred because they counter CNI afferent vasoconstriction; tight BP control protects the graft.

BOARD TRAP

In CNI-related HTN with graft concerns, a CCB is favored — avoid ACEi/ARB initially if it unmasks renal artery stenosis or raises creatinine.

3. Transplant renal artery stenosis (RAS)

WHAT YOU SEE

Renal-artery stenosis trap, severe HTN

WHAT IT ENCODES

Transplant RAS presents as refractory HTN, graft dysfunction, and a creatinine RISE after starting ACEi/ARB. Suspect with resistant HTN 3–24 months post-transplant; confirm with Doppler/angiography.

BOARD TRAP

A creatinine jump after starting an ACEi/ARB post-transplant = transplant renal artery stenosis, not just 'expected ACEi effect' — image the graft artery.

4. Skin cancer — #1 malignancy

WHAT YOU SEE

Skin cancer field, SCC > BCC

WHAT IT ENCODES

Skin cancer is the most common post-transplant malignancy, and squamous cell carcinoma (SCC) outnumbers basal cell — the reverse of the general population. Counsel sun protection and surveillance.

BOARD TRAP

Post-transplant, SCC > BCC (ratio flips). For recurrent skin cancers, switching to an mTOR inhibitor (sirolimus) reduces risk.

5. Sirolimus reduces skin-cancer recurrence

WHAT YOU SEE

Consider sirolimus note in skin field

WHAT IT ENCODES

mTOR inhibitors (sirolimus/everolimus) have antineoplastic properties; switching from a CNI to an mTOR inhibitor lowers recurrent skin cancer and Kaposi risk.

BOARD TRAP

For recurrent post-transplant skin cancer, switch toward sirolimus — don't just intensify CNI, which can worsen malignancy risk.

6. PTLD — EBV-driven lymphoma

WHAT YOU SEE

PTLD lymph-node forest, EBV trees

WHAT IT ENCODES

Post-transplant lymphoproliferative disorder is EBV-driven (especially EBV-naïve recipients of EBV+ organs). First step is REDUCING immunosuppression; add rituximab for CD20+ disease ± chemo.

BOARD TRAP

PTLD is EBV-associated lymphoma — first move is reduce immunosuppression, not immediately chemo; CMV is not the driver here.

7. BK virus nephropathy → graft dysfunction

WHAT YOU SEE

BK virus castle, BACK OFF immunosuppression

WHAT IT ENCODES

BK polyomavirus causes interstitial nephritis/graft dysfunction; biopsy shows tubular cytopathic changes/SV40+ staining. Management is BACKING OFF immunosuppression — no reliable antiviral.

BOARD TRAP

BK nephropathy mimics rejection on biopsy but treatment is OPPOSITE — reduce immunosuppression, not increase it; antivirals don't work.

8. Checkpoint inhibitor trap

WHAT YOU SEE

Checkpoint-inhibitor trap, wakes T-cells

WHAT IT ENCODES

Immune checkpoint inhibitors (anti-PD1/CTLA4) can precipitate acute allograft rejection by reactivating T-cells — a high-stakes interaction when a transplant patient develops cancer needing immunotherapy.

BOARD TRAP

Checkpoint inhibitors risk triggering graft rejection — they are not a benign cancer therapy in transplant recipients.

9. Persistent hyperparathyroidism / hypercalcemia

WHAT YOU SEE

Bone & calcium wing, tertiary HPT

WHAT IT ENCODES

Tertiary hyperparathyroidism can persist after transplant, causing hypercalcemia and hypophosphatemia as autonomous glands keep secreting PTH despite restored renal function.

BOARD TRAP

Post-transplant hypercalcemia is often persistent (tertiary) HPT — not new primary disease; may need cinacalcet or parathyroidectomy.

10. Steroid → hip pain → avascular necrosis

WHAT YOU SEE

Steroids + hip pain → AVN

WHAT IT ENCODES

Chronic glucocorticoids cause avascular (osteonecrosis) of the femoral head — new hip/groin pain in a steroid-treated transplant patient warrants MRI for AVN, plus osteoporosis risk.

BOARD TRAP

New atraumatic hip pain on chronic steroids = avascular necrosis (MRI), not just osteoarthritis — image early before collapse.

11. Post-transplant diabetes (NODAT) — CNI/steroids

WHAT YOU SEE

Diabetes in CV tower, steroid/tacrolimus link

WHAT IT ENCODES

New-onset diabetes after transplant is driven by steroids and tacrolimus (> cyclosporine). It raises CV risk; manage with glucose control and possibly switching toward cyclosporine.

BOARD TRAP

Tacrolimus causes MORE post-transplant diabetes than cyclosporine — the CNI choice matters; steroids compound it.

12. Anemia — marrow suppression / residual CKD

WHAT YOU SEE

Anemia corner, marrow suppression AZA/MMF

WHAT IT ENCODES

Post-transplant anemia comes from marrow-suppressive drugs (azathioprine, MMF, valganciclovir), residual CKD with low EPO, and iron deficiency. Work through drug effects before EPO.

BOARD TRAP

New cytopenia is often drug-induced (AZA/MMF/valganciclovir), not graft failure — review meds before attributing to CKD anemia.

13. PJP prophylaxis — TMP-SMX × 6 months

WHAT YOU SEE

PJP prophylaxis sentry, 6-month marker

WHAT IT ENCODES

Pneumocystis jirovecii prophylaxis with TMP-SMX is given for ~6–12 months post-transplant (also covers Toxoplasma, Nocardia, UTIs). Restart with rejection-treatment intensification.

BOARD TRAP

PJP prophylaxis is TMP-SMX (not just for HIV) — stopping early or during augmented immunosuppression risks PJP pneumonia.

14. Boards decision band

WHAT YOU SEE

Bottom band: CV death, skin cancer #1, PTLD=EBV, BK back off

WHAT IT ENCODES

High-yield late algorithm: after year 1 CV disease kills; skin cancer (SCC) is #1 malignancy; PTLD=EBV→reduce immunosuppression; BK→reduce immunosuppression; resistant HTN 3–24 mo → think RAS; PJP prophylaxis TMP-SMX ×6 mo.

BOARD TRAP

The repeated pattern: viral/lymphoproliferative complications (BK, PTLD) are managed by LOWERING immunosuppression, opposite to rejection.